Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CE33]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2021

STRENGTH OF MATERIALS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Derive the Relationship between the Elastic Constants E,G,K **07**

b. A compound bar shown in Fig. 1. consists of a 30 mm diameter steel rob encased in a copper tube of internal diameter 30mm and external diameter 40mm. find the stresses produced in steel and copper rod when a load of 100 N falls from a height of 40 mm. take $E_s = 2 \times 10^5$ N/mm^2, $E_c = 1 \times 10^5$ N/mm^2 **07**

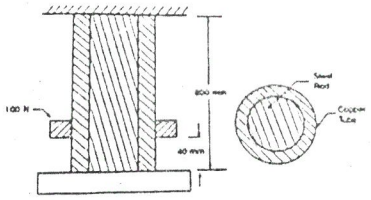

Fig 1

2. a. Draw shear force and bending moment for the beam shown in Fig. 2. **10**

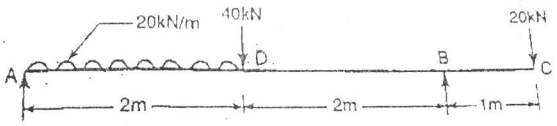

Fig 2

b. Derive the relationship between load, shear force and bending moment. **04**

3. a. Derive the bending equation M/I = f/Y =E/R, stating the assumptions **07**

b. A circular pipe of external diameter 70mm and thickness 8mm is used as a simply supported beam over an effective span of 3 m. find the maximum concentrated load that the beam can carry at the center of the span if permissible stresses in the tube are150n/mm^2 **07**

OR

4. a. The unsymmetrical I section having sectional dimensions as, top flange : 80 X 20mm ,web :10 X 200 mm & bottom flange : 160 X 20mm, is subjected to shear force of 40 kN. Find the shear stress at all salient points and draw the stress diagram. **09**

b. Prove that the maximum shear stress for a rectangular section is 1.5 times the average shear stress **05**

5. a. Derive the equation for torsion $T/J = q/r = G\theta/L$ stating the assumptions **09**

b. A hollow circular shaft 200mm external diameter and metal thickness 25mm is transmitting power at 200rpm. The angle of twist over a length of 2m was found to be 0.5 degrees. Calculate the power transmitted and maximum shear stress induced in the section. Take modulus of rigidity as $84kN/mm^2$ **05**

OR

6. a. A hollow propeller shaft of a steam ship is to transmit 3750kW at 240rpm. If the internal diameter 0.8 times the external diameter and if maximum shear stress developed is to be limited to $160N/mm^2$, determine the size of the shaft. **07**

b. A hollow circular shaft of 6m length and inner and outer diameters of 75mm and 100mm is subjected to a torque of 10kNm. If G= 80 GPa, determine the maximum shear stress produced and the total angle of the twist. **07**

7. a. Derive lame's equation for radial and hoop stresses for thick cylinders subjected to internal and external fluid pressure. **05**

b. A thick cylindrical pipe with outside diameter 300mm and internal diameter 200mm is subjected to an internal fluid pressure of $14N/mm^2$ .determine the maximum hoop stress across the thickness of the pipe. **09**

OR

8. a. Write the stress tensor for stresses in three dimensional system **05**

b. The direct stresses acting at a point in a strained material are shown in fig 4. Find the normal, tangential and the resultant forces on a plane 30^0 to the plane of the major principal stress. Find the obliquity of the resultant stress also. **09**

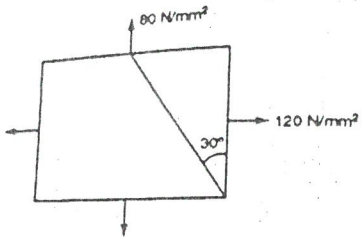

Fig 4